

Overall Review

IEEE Symposium on Security and Privacy: Bridging Academic Rigour and Industrial Deployment (2022-2024)

Research at the IEEE Symposium on Security and Privacy (S&P) from 2022 through 2024 reveals a persistent disconnect between theoretically sophisticated adversarial machine learning methodologies and the practical constraints governing industrial deployment. Unlike work emphasising individual attack sophistication or isolated defensive mechanisms, IEEE S&P contributions expose fundamental tensions between computational efficiency, realistic threat modelling, and the operational requirements of production systems. This body of work demonstrates that bridging the research-practice gap requires not only technically sound solutions but also careful consideration of resource constraints, regulatory compliance, and the adversarial evolution that occurs when defences meet adaptive attackers.

Inappropriate Threat Modeling and Resource Assumptions

A fundamental contributor to the research-practice gap lies in academic research operating under unrealistic assumptions about adversary capabilities and computational resources. Many advanced adversarial attacks and defenses presume white-box access to model internals—knowledge of architecture, parameters, and training procedures—that rarely exists in practical attack scenarios against commercial systems (Wenger et al., 2023; Li et al., 2023). This mismatch extends to computational overhead: established property inference attacks typically require training hundreds or thousands of shadow models to approximate victim model behavior, incurring massive computational costs that render these attacks impractical for resource-constrained adversaries (Chaudhari et al., 2023).

Recent work directly addresses these efficiency barriers. The SNAP attack achieves high success rates in property inference while being up to 56.5 times faster than prior methods by using poisoning strategies that require only a few shadow models rather than hundreds (Chaudhari et al., 2023). Similarly, query-efficient attacks like BounceAttack optimize for decision-based black-box settings where attackers receive only final predictions, reducing the number of queries needed to successfully craft adversarial examples against commercial systems (Wan et al., 2024). For Automatic Speech Recognition platforms deployed commercially, ALIF tackles both the cost and fragility of prior black-box attacks by demonstrating significant improvements in query efficiency and resilience to model updates that occur in production environments (Cheng et al., 2024). These advances reflect a critical shift toward attacks that operate within the resource and access constraints faced by real adversaries.

Evaluation Metrics and Privacy as Non-Average-Case Properties

Academic evaluations frequently employ inappropriate metrics that fail to capture true security risks in deployed systems. Membership inference attacks, for instance, are commonly evaluated using average-case metrics like overall accuracy or area under the ROC curve. However, privacy is fundamentally a worst-case concern: a defense succeeds only if it protects all individuals, not just the majority. Proper evaluation should therefore measure true-positive rate at low false-positive rates, as this reflects the attacker's ability to reliably identify specific victims while maintaining stealth (Carlini et al., 2022). This methodological flaw leads to overestimation of defense effectiveness and underestimation of real privacy risks.

The accuracy-privacy trade-off further complicates practical deployment. Deep ensemble models designed to maximize prediction accuracy inherently increase vulnerability to membership inference attacks—achieving high model quality directly conflicts with maintaining privacy guarantees (Rezaei et al., 2023). Even supposedly robust privacy measures prove insufficient in practice: Differential Privacy mechanisms fail to fully mitigate leakage of Personally Identifiable Information from language models without aggressive parameter tuning that severely degrades model utility (Lukas et al., 2023). Moreover, theoretical privacy guarantees can be undermined entirely by implementation weaknesses. The Gaussian mechanism for Differential Privacy suffers from floating-point representation attacks, while proposed discrete alternatives remain vulnerable to timing side-channel attacks that reveal the magnitude of sampled noise, completely invalidating privacy guarantees (Jin et al., 2022).

Scalability Challenges in Certified Robustness

The pursuit of formal robustness guarantees reveals a fundamental scalability barrier that prevents deployment of academically rigorous defenses. Complete verification techniques provide strong, mathematically proven guarantees about model behavior under adversarial perturbations but fail to scale beyond toy datasets like CIFAR-10 to industrial-sized deep neural networks (Li et al., 2023). Conversely, probabilistic verification methods that do scale to realistic model sizes often provide bounds so loose as to be trivial or useless for practical security decisions. This creates an intractable dilemma: defenses with meaningful guarantees cannot be deployed at scale, while scalable defenses cannot provide meaningful guarantees.

Recent work attempts to resolve aspects of this trade-off for specific domains. Text-CRS advances certified robustness in natural language processing by providing mathematical guarantees not merely for simple word substitution—the focus of prior research—but for practical word-level operations including insertion and deletion, significantly expanding the scope of certifiable defense mechanisms (Zhang et al., 2024). However, these domain-specific advances do not resolve the fundamental computational barriers preventing widespread deployment of certified robustness across diverse application areas.

Efficiency Constraints in Distributed Learning

Federated Learning deployments introduce resource constraints that render many academic defenses impractical. Traditional approaches to recovering from poisoning attacks require "train-from-scratch" remediation—retraining the entire model from initialization—which imposes massive computational and communication burdens on resource-constrained client devices such as smartphones or IoT sensors (Cao et al., 2023). Solutions like FedRecover address this efficiency gap by restoring model accuracy

through estimation of historical client updates rather than complete retraining, making remediation feasible within realistic device constraints (Cao et al., 2023).

The LOKI attack exposes critical weaknesses in industry-standard defenses when applied to practical, large-scale federated deployments. Secure aggregation mechanisms designed to protect client privacy prove vulnerable when deployed in realistic FedAVG settings with many participants, undermining the fundamental privacy premise of cross-device federated learning systems (Zhao et al., 2024). This vulnerability is particularly concerning because secure aggregation has been widely adopted as a standard defense in commercial federated learning platforms, suggesting that industrial deployment practices may rely on security guarantees that do not hold under realistic threat models.

Vertical Federated Learning—commonly deployed in applications like financial fraud detection where multiple institutions collaboratively train models on complementary features—presents unique practical challenges. The BadVFL framework reveals that adversaries can successfully inject backdoors despite lacking access to labels or the full feature space, constraints that fundamentally change attack dynamics compared to horizontal federated learning (Naseri et al., 2024). Similarly, Adversarial Dominating Inputs attacks demonstrate that adversaries can achieve complete control over joint inference in Vertical FL using minimal external data, undermining the proportional reward mechanisms that incentivize honest participation in collaborative training (Pang et al., 2023).

Stealthy and Adaptive Attack Strategies

Practical deployment environments incentivize adversarial strategies fundamentally different from those studied in controlled academic settings. Attackers must evade detection by defensive mechanisms that often employ multiple layers of protection rather than single, isolated defenses. The 3DFed framework integrates adaptive evasion strategies specifically designed to counter multilayered defenses in black-box settings, challenging the unrealistic assumption that defenders utilize only simple, single-objective detection heuristics (Li et al., 2023).

Attack stealth has evolved beyond simply minimizing perturbation magnitude. The Jigsaw Puzzle attack achieves evasion by selectively targeting small malicious subpopulations rather than entire classes, exploiting the fact that traditional backdoor defenses assume attackers compromise all samples of a target class—a premise violated in realistic targeted attack scenarios (Yang et al., 2023). Similarly, the BAGS methodology demonstrates that sophisticated backdoor attacks require minimal poisoning ratios when attackers focus on efficient sample selection early in training, making detection through statistical anomaly analysis substantially more difficult (Han et al., 2024). In multimodal learning systems, data efficiency becomes critical: successful attacks must account for modality interactions while minimizing computational overhead and the number of poisoned samples needed to achieve attack objectives.

Evolving Attack Surfaces in Production Systems

As machine learning systems mature and diversify, new attack surfaces emerge that academic research has not traditionally considered. Core model components increasingly become attack targets: BadEncoder demonstrates that adversaries can compromise pre-trained encoders used in self-supervised learning, ensuring that all downstream classifiers trained on these encoders inherit malicious behaviors regardless of subsequent fine-tuning (Jia et al., 2022). This supply chain attack

model poses severe risks for industry practices that rely on shared pre-trained models to reduce computational costs.

Physical and hardware-level attacks reveal vulnerabilities in deployment infrastructure rather than algorithmic weaknesses. The DeepSteal framework demonstrates remote theft of deep neural network weights from Machine Learning as a Service platforms by leveraging Rowhammer-based memory side-channel attacks (HammerLeak), achieving high-accuracy model extraction through partial leakage of Most Significant Bits that guide substitute model training (Rakin et al., 2022). At the hardware accelerator level, DeepShuffle addresses adversarial fault injection attacks that exploit predictable, fixed data transmission patterns in commercial cloud-FPGA accelerators like Xilinx DPU, demonstrating that even hardware-level design decisions create exploitable attack surfaces (Luo et al., 2024).

Emerging deployment techniques introduce novel vulnerabilities. Test-Time Adaptation models designed to enhance robustness in safety-critical applications prove vulnerable to test-time poisoning attacks (TePA), revealing that adaptive mechanisms intended to improve reliability can themselves become attack vectors when adversaries manipulate the adaptation process (Cong et al., 2024). Similarly, input encoding attacks exploit the gap between machine text processing and human perception by using invisible characters or homoglyphs to create imperceptible perturbations, successfully compromising commercial Natural Language Processing services including content filtering and search indexing systems (Boucher et al., 2022).

Infrastructure and Deployment-Specific Vulnerabilities

Real-world ML systems operate within complex deployment pipelines that introduce vulnerabilities absent in isolated academic settings. Privacy breaches extend beyond training data to graph-structured data relationships: the LINKTELLER attack performs link re-identification on inductive Graph Neural Networks through node influence analysis via black-box queries, proving effective even when attackers lack prior knowledge of full graph structure—a realistic scenario in many social network and recommendation system deployments (Shen et al., 2022).

Physical authentication systems face practical attacks that exploit implementation shortcuts rather than cryptographic weaknesses. DepthFake spoofs 3D face authentication systems using only 2D photos by exploiting fixed scatter patterns in commercial depth sensors, revealing that cost-saving hardware design decisions create fundamental security vulnerabilities (Wu et al., 2023). The ImU attack demonstrates physical impersonation through consistent and natural style changes across multiple poses, achieving high success rates even against sophisticated facial recognition systems deployed in access control applications (An et al., 2023).

The security of development tools themselves proves problematic. Systematic investigation of code generated by GitHub Copilot revealed that approximately 40% of generated programs contain cybersecurity weaknesses classified as Common Weakness Enumerations, suggesting that machine learning-assisted development tools perpetuate insecure patterns from open-source training data into industry codebases (Pearce et al., 2022). This finding has profound implications for software supply chain security as AI-assisted coding tools gain widespread adoption.

Defensive Inadequacies and Natural Vulnerabilities

Traditional copyright protection mechanisms for machine learning models prove fundamentally inadequate against realistic threats. Watermarking techniques are often invasive—degrading model performance—and easily defeated by model extraction attacks that can remove or override watermarks (Chen et al., 2022). Alternative approaches like DEEPJUDGE propose non-invasive testing frameworks using multi-level metrics to characterize model similarity and provide ownership evidence against fine-tuning, pruning, and extraction threats, though these methods sacrifice the definitiveness of cryptographic watermarking (Chen et al., 2022).

Beyond intentionally injected backdoors, research reveals that normally trained models contain natural backdoors—inherent vulnerabilities arising from the training process itself rather than malicious intervention. Model Orthogonalization (MOTH) addresses this pervasive issue by substantially enlarging class distances to make models resilient against both injected and native backdoors, representing a shift from reactive defense against known attacks toward proactive hardening against fundamental model weaknesses (Tao et al., 2022). Similarly, poisoning attacks have been repurposed not merely to degrade model accuracy but to actively boost information leakage, demonstrating that models trained with Differential Privacy-SGD remain susceptible to property inference attacks even at tight privacy budgets (Mahloujifar et al., 2022).

Post-training detection mechanisms must operate within realistic constraints that limit their applicability. MM-BD addresses the practical need for backdoor detection without requiring access to original training data or assuming knowledge of specific attack patterns, offering a solution deployable by downstream model users who lack visibility into training procedures (Wang et al., 2024). However, such detection methods inherently face the challenge of adaptive adversaries who design attacks specifically to evade known detection mechanisms.

Deployment-Driven Security Requirements

Practical deployment introduces requirements beyond pure attack resistance. In Trusted Execution Environment (TEE)-shielded deployments, TEESLICE demonstrates that many proposed research solutions fail to meet essential performance demands like low latency while simultaneously proving vulnerable to knowledgeable attackers, exposing a fundamental utility-security trade-off in secure hardware deployment scenarios (Zhang et al., 2024). Regulatory compliance creates additional constraints: Vertical Data Minimization (vDM) explicitly addresses GDPR requirements by providing mechanisms that minimize data collection while preserving model utility, reflecting how privacy regulations increasingly drive technical requirements for production systems (Staab et al., 2024).

Cross-device federated learning deployments exhibit robustness characteristics absent from academic threat models. Empirical studies reveal that cross-device Federated Learning remains highly robust even with simple defenses when realistic, low attack rates are employed—contradicting much academic literature that assumes unrealistically high compromise rates among clients (Shejwalkar et al., 2022). This finding suggests that research efforts must recalibrate assumptions about attacker prevalence to offer meaningful insights for practical robust algorithm design.

Usability considerations critically impact security deployment decisions. On Augmented and Virtual Reality platforms, user authentication proves "overwhelmingly unusable" due to difficulty entering passwords via virtual keyboards, driving users toward insecure practices (Stephenson et al., 2022). This usability barrier demands that new authentication mechanisms gain operating system support to

become viable alternatives, illustrating how human factors and platform ecosystem constraints shape which security technologies can successfully deploy regardless of their technical merit.

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Yearly Breakdown

1.) 2022

Adversarial machine learning faces a fundamental challenge: the gap between theoretical research and practical industry security needs. This disparity is exploited by increasingly sophisticated attacks that subvert production systems. Academic evaluations often fail to capture real-world risks. For example, membership inference attacks (MIA) should be evaluated by their true-positive rate (TPR) at low false-positive rates (FPR), since privacy is not an average-case metric (Carlini et al., N/A).

Evolving Attack Landscape and Deployment Vulnerabilities

Adversaries are actively targeting fundamental ML components and the unique constraints of deployed platforms:

- **Core Model Corruption:** Attacks now strike deep into the ML pipeline, targeting foundational, pre-trained components. **BadEncoder** attacks compromise pre-trained encoders used in self-supervised learning (SSL), ensuring downstream classifiers inherit malicious behaviour (Jia et al., 2022).
- **Physical and Remote Exploits:** The threat extends to hardware infrastructure. The **DeepSteal** framework demonstrates remote theft of Deep Neural Network (DNN) weights by leveraging a **Rowhammer-based memory side-channel attack (HammerLeak)**, posing a direct threat to Machine Learning as a Service (MLaaS) platforms (Rakin et al., N/A). This method achieves high accuracy extraction by using partially leaked Most Significant Bits (MSBs) to guide substitute model training (Rakin et al., N/A).
- **Graph Data Breaches:** Privacy vulnerabilities are pronounced in graph-structured data. The **LINKTELLER** attack performs link re-identification on **inductive Graph Neural Networks (GNNs)** by

using node influence analysis via black-box queries, proving effective even when the attacker lacks prior knowledge of the full graph structure (Shen et al., N/A).

- **Stealthy Input Attacks:** Moving beyond conventional adversarial perturbations, attacks now exploit underlying text encodings. **Imperceptible encoding-based perturbations** exploit the gap between how machines process text and how humans perceive it (e.g., using invisible characters or homoglyphs), enabling black-box attacks against commercial Natural Language Processing (NLP) services and compromising applications like content filtering and search indexing (Boucher et al., N/A).
- **Insecure Tooling:** Even modern developer tools are vulnerable. A systematic investigation into code generated by tools like **GitHub Copilot** revealed that approximately **40%** of generated programs contained cybersecurity weaknesses (Common Weakness Enumerations or CWEs), suggesting that insecure patterns from open-source code are perpetuated in industry tools (Pearce et al., N/A).

Deficiencies in Defensive Mechanisms

Academic defences and guarantees often prove insufficient or impractical when deployed:

- **DP Implementation Flaws:** Theoretical safeguards like Differential Privacy (DP) are undermined by implementation weaknesses. The **Gaussian mechanism** suffers from floating-point representation attacks, and proposed discrete counterparts (designed to mitigate floating-point flaws) are vulnerable to **timing side-channel attacks** that reveal the magnitude of the sampled noise, invalidating DP guarantees (Jin et al., N/A).
- **IP Protection Failures:** Traditional copyright protection methods, such as **watermarking**, are fundamentally flawed—often invasive and easily defeated by model extraction attacks (Chen et al., N/A). Countermeasures like **DEEPJUDGE** propose a non-invasive testing framework using multi-level metrics to characterize model similarity and provide evidence for ownership claims against fine-tuning, pruning, and extraction threats (Chen et al., N/A).
- **Targeted Poisoning:** Poisoning attacks have been repurposed not just to harm model accuracy, but to **boost information leakage**. Models trained with DP-SGD are susceptible to property inference attacks even at tight privacy budgets, highlighting that DP alone is inadequate protection against malicious participants (Mahloujifar et al., N/A).
- **Model Hardening:** The pervasive issue of **natural backdoors** (inherent vulnerabilities in normally trained models) requires active hardening techniques like **Model Orthogonalization (MOTH)**, which substantially enlarges class distances to make models resilient to both injected and native backdoors (Tao et al., N/A).
- **Usability Barriers:** On platforms like Augmented and Virtual Reality (AR/VR), user authentication is **"overwhelmingly unusable"** due to the difficulty of entering passwords via virtual keyboards (Stephenson et al., N/A). This usability challenge drives users toward insecure practices and demands that new authentication mechanisms gain OS support to become viable alternatives (Stephenson et al., N/A).
- **Federated Learning Realism:** Studies show that cross-device Federated Learning (FL) is actually highly robust even with simple defenses when realistic, low attack rates are employed—contrary to

much academic literature assuming unrealistic compromise rates (Shejwalkar et al., N/A). This implies that research efforts must pivot toward practical constraints to offer meaningful insights and useful robust algorithms (Shejwalkar et al., N/A).

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2.) 2023

The gap between adversarial machine learning (AML) research and industry deployment stems from theoretical complexity that fails to reflect real-world constraints, particularly regarding attacker capabilities and defensive feasibility. Many advanced adversarial attacks and defenses assume unrealistic **white-box access** to model internals or rely on computationally prohibitive methods (Wenger et al., 2023; Li et al., 2023). For instance, established property inference attacks typically require training hundreds or thousands of shadow models, incurring massive computational overhead (Chaudhari et al., 2023). Attacks like SNAP address this efficiency gap by using poisoning to achieve high success rates while being significantly faster (up to 56.5x) and relying on only a few shadow models (Chaudhari et al., 2023). Similarly, new backdoor attacks focus on enhanced stealth and realistic constraints. The Jigsaw Puzzle (JP) attack exploits the impractical assumption that backdoors target an entire class, instead selectively targeting small malicious subpopulations to evade detection (Yang et al., 2023). Furthermore, sophisticated Federated Learning (FL) attacks now operate successfully in challenging **black-box settings** against multi-layered defenses, eliminating the need for white-box knowledge previously assumed necessary (Li et al., 2023). Physical attacks, such as

DepthFake against 3D face authentication systems, reveal fundamental flaws in industry solutions by exploiting practical vulnerabilities like fixed scatter patterns used by depth sensors, demonstrating how theoretical assumptions of system integrity can be practically broken (Wu et al., 2023).

Defences often struggle with practicality due to efficiency concerns, resource constraints on client devices, or utility degradation from overly aggressive security measures. For example, in FL, recovering a compromised model using traditional "train-from-scratch" methods is impractical due to the massive computation and communication burden on resource-constrained clients (Cao et al., 2023). Solutions like FedRecover mitigate this gap by efficiently restoring models through estimating historical updates, reducing client load (Cao et al., 2023). Theoretical certified robustness, while desirable for strong guarantees, faces inherent trade-offs: highly accurate formal verification approaches (complete verification) do not scale to industrial-sized deep neural networks, while scalable methods (probabilistic verification) often provide loose or trivial robustness bounds (Li et al., 2023). These defensive limitations create a challenging **accuracy-privacy trade-off**, especially in ensemble models, where achieving high model accuracy inherently increases vulnerability to membership inference attacks (Rezaei et al., 2023). Moreover, traditional defenses like Differential Privacy (DP) often fail to fully mitigate privacy leakage of Personally Identifiable Information (PII) from language models, requiring overly aggressive scrubbing that severely degrades model utility (Lukas et al., 2023). Addressing these deficiencies requires efficient unlearning solutions that succeed even when external information (like the trigger) is unavailable (Zhu et al., 2023), or solutions that target real-world infrastructure vulnerabilities, such as implementing public integrity verification against potentially dishonest service providers who control run-time models (Wang et al., 2023).

The gap between adversarial machine learning (AML) research and industry practice is defined by the tension between theoretically rigorous yet often impractical defensive measures and the need for efficient, real-world solutions (Li et al., 2023; Lukas et al., 2023). Robustness certification exemplifies this gap: techniques providing strong, formal guarantees (complete verification) suffer from the **scalability challenge** and cannot be applied to deep neural networks of common industrial sizes, such as those beyond CIFAR-10 scale. Consequently, only looser verification methods are tractable for larger models. Similarly, addressing practical operational constraints poses significant barriers for traditional defences. Removing a poisoning attack in Federated Learning (FL) typically necessitates expensive "train-from-scratch" recovery methods, incurring substantial computational and communication costs intolerable for resource-constrained client devices (Cao et al., 2022).

Approaches like FedRecover mitigate this efficiency constraint by restoring model accuracy through estimating client updates from historical data. Furthermore, achieving high model quality often conflicts with maintaining privacy, creating a fundamental **accuracy-privacy trade-off**; deep ensemble models designed for accuracy exhibit increased vulnerability to membership inference attacks (Rezaei et al., 2023). Even standard privacy measures like Differential Privacy (DP) are compromised in practice, as aggressive application of DP or scrubbing is necessary to limit Personally Identifiable Information (PII) leakage, severely degrading the language model's utility (Lukas et al., 2023).

In contrast, recent offensive research highlights the push toward realism by challenging traditional academic threat models that assume impractical resources or knowledge (Li et al., 2023). Traditional property inference attacks, for example, relied on attackers training hundreds or thousands of shadow models—a massive computational overhead. Attacks like SNAP circumvent this by using efficient poisoning and distinguishing tests that require only a small number of shadow models (at most four)

and achieve significantly higher speed and success rates than prior work, moving AML research closer to practical feasibility (Chaudhari et al., 2023). In critical areas like Vertical Federated Learning (VFL), threats such as Adversarial Dominating Inputs (ADIs) demonstrate success in black-box settings, achieving complete control over joint inference using minimal external data and undermining the proportional rewards system that incentivises participation (Pang et al., 2023). Furthermore, attacks are becoming covert and architecture-aware to evade industry-standard multi-layered defenses; the 3DFed framework integrates adaptive evasion strategies in a black-box setting, countering the unrealistic assumption that defenders utilize only simple, single-objective heuristics (Li et al., 2023). Similarly, the Jigsaw Puzzle attack achieves stealth by selectively targeting small malicious subpopulations rather than the entire class, evading detection mechanisms designed around the premise of universal class compromise (Yang et al., 2023). These constraints become acutely visible in physical attacks: DepthFake exposes vulnerabilities in 3D face authentication systems by exploiting simple hardware flaws like fixed scatter patterns using only a 2D photo (Wu et al., 2023), while the ImU attack demonstrates physical impersonation using consistent and natural style changes across multiple poses, achieving high success even against sophisticated recognition systems (An et al., 2023).

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The gap between theoretical adversarial machine learning (AML) research and practical industry application remains a crucial focus, encompassing challenges related to attack efficiency, deployment security, and the necessity for generalized robustness across complex, real-world systems. Several recent papers highlight this gap by proposing novel methodologies tailored to real-world constraints or by revealing vulnerabilities in emerging industry defense standards. For instance, addressing the need for cost-efficient adversarial attacks in black-box environments, research has introduced methods like **BounceAttack** (Wan et al., 2024), which emphasizes query efficiency in decision-based black-box settings. Similarly, the **ALIF** framework (Cheng et al., 2024) specifically tackles the cost and fragility of prior black-box attacks on commercial Automatic Speech Recognition (ASR) platforms by demonstrating significant improvements in query efficiency and resilience to model updates in the wild. In complex multimodal systems, where attack efficiency is complicated by modality interactions, the **BAGS** methodology (Han et al., 2024) provides a data and computation-efficient solution for backdooring, demonstrating that highly sophisticated attacks require minimal poisoning ratios by focusing on efficient sample selection early in training.

On the defensive side, security research targets the practical vulnerabilities introduced by current enterprise and deployment practices. In Federated Learning (FL), the **LOKI** attack (Zhao et al., 2024) exposes critical weaknesses in industry-standard defenses like secure aggregation (SA) when applied in the practical, large-scale FedAVG setting, undermining the premise of privacy in cross-device deployments. Conversely, the **BadVFL** framework (Naseri et al., 2024) explores backdoor attacks unique to Vertical FL (VFL) deployments—common in applications like financial fraud detection—where the adversary must succeed despite lacking access to labels or the full feature space, a non-trivial practical challenge. Furthermore, vulnerabilities extend to the hardware level: the **DeepShuffle** framework (Luo et al., 2024) addresses adversarial fault injection attacks that exploit the predictable, fixed data transmission patterns within commercial cloud-FPGA accelerators (like Xilinx DPU), promoting dynamic change as a viable defense. Even in the context of emerging deployment techniques, the **TePA** approach (Cong et al., 2024) identifies an impactful new attack surface against Test-Time Adaptation (TTA) models used in safety-critical applications, demonstrating that TTA methods are currently vulnerable to test-time poisoning despite their intended use in enhancing robustness.

Finally, the focus of the field is shifting towards providing robust, universally applicable solutions that can operate within realistic system constraints and maintain regulatory compliance. **MM-BD** (Wang et al., 2024) addresses the practical need for post-training backdoor detection without relying on access to the original training set or assuming a specific attack pattern, offering a model easily applicable by downstream users. Similarly, the **TEESLICE** solution (Zhang et al., 2024) directly confronts the utility-security trade-off in TEE-shielded deployment, proving that many proposed research solutions are vulnerable to knowledgeable attackers and fail to meet essential performance demands like low latency. **Text-CRS** (Zhang et al., 2024) advances certified robustness in NLP by creating a framework

that provides mathematical guarantees not just for simple word substitution (as in prior research), but for practical word-level operations like insertion and deletion, significantly expanding the scope of certifiable defense. This movement towards real-world applicability is underscored by frameworks like Vertical Data Minimization (vDM) (Staab et al., 2024), which is fundamentally driven by compliance regulations such as GDPR, explicitly demanding mechanisms that minimize data collection while preserving utility in deployment.

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